

CPSC 121 – Lab 01: Problem Solving

Programming Problems:

Complete the following programming problems. Ensure you have demonstrated all of them to the instructor and compress all of the files into a .zip file named `Lab01_LastName_FirstName.zip`. Attach the zip file to the Assignment and answer the questions on the Test “Lab 01: Problem Solving Part 1” as well.

1.

Create a revised version of the `Lincoln` application (given in the .zip file) such that double-quotes appear around the quotation.

2.

Write an application (named `Average`) that uses three `int`'s (variables declared and assigned in your code) and prints their average as a `double` data type. **Test – try 2, 3, 5 and you should get 3.3333333.**

3.

Write an application (named `Computations`) that uses two `double` numbers (variables declared and assigned in your code) and prints their sum, difference and product.

4.

Create a version of the `TemperatureConverter` application (given in the .zip file) to convert from Fahrenheit to Celsius. The Fahrenheit temperature should be declared and assigned in your code. **Test – try a Fahrenheit temperature of 98.6 and you should get 37.0 or nearest equivalent.**

5.

Write an application that converts miles to kilometers (named `MilesToKilometers`). (One mile equals 1.60935 kilometers.) The miles value should be declared and assigned in your code as a `double` value. Note – conversion factors make great constants! Print the miles and equivalent kilometers.

Rubric

- ❖ Do the programs work as described? (Needs my sign-off): **10 points**
- ❖ **Total: 10 points**

Submitting the Lab

Compress the submission files (see below) into a .zip file named: **Lab01_LastName_FirstName.zip** When you are done and ready to submit, submit the following files in the .zip file:

- Lincoln.java
- Average.java
- Computations.java
- TemperatureConverter.java
- MilesToKilometers.java